Hyperbolic Flavor Geometry:
Mixing, CP Violation, and Fermion Masses from Arithmetic Hyperbolic 3-Manifolds

Marvin L. Gentry[1]

[1]Independent Researcher, Seattle WA; `drmlgentry@protonmail.com`; ORCID: 0009-0006-4550-2663

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Abstract

We study two compact arithmetic hyperbolic 3-manifolds: $M_{\text{PMNS}} = m003(-2,3)$ (`OrientableClosedCensus[1` volume 0.9814, $H_1 = \mathbb{Z}/5$) and $M_{\text{CKM}} = m006(-5,2)$ (`OrientableClosedCensus[43]`, volume 2.0289, $H_1 = \mathbb{Z}/5$). M_{PMNS} is the Meyerhoff manifold, the unique minimum-volume closed orientable hyperbolic 3-manifold [3].

We prove three geometric theorems independent of any physical interpretation. (I) The cusp of the ancestor $m003$ has shape exactly $\tau = e^{i\pi/3} = \omega$ (the Eisenstein unit), as a consequence of the trace field $\mathbb{Q}(\sqrt{-3})$. The surgery formula $|H_1(m003(p,q))| = 5|2p+q|$, proved from the fundamental group presentation, gives $H_1(M_{\text{PMNS}}) = \mathbb{Z}/5$ as a theorem. (II) The invariant trace fields $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary) and $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (real) place the manifolds in distinct commensurability classes; the discriminant sign distinguishes twisted loxodromic from purely hyperbolic holonomy. (III) The unique connected degree-5 covering space of M_{PMNS} introduces new torsion prime $11 = L_5$ (fifth Lucas number), equal to the Farey intersection $\det[(-2,3),(-5,2)] = 11$ of the two filling slopes.

Guided by these rigid geometric results, we then present computational observations suggesting nontrivial numerical correspondences with Standard Model flavor parameters. No continuously tuned physical fit parameters enter once the word triples are fixed: the PMNS mixing matrix arises from the Borel factor of the Iwasawa decomposition (Frobenius fitness 0.005087, minimum of the construction within the scanned word ensemble, $p < 10^{-4}$ against 50,000 Haar-random unitaries); the leptonic CP phase is $\delta = \pi + \varphi(\text{aaB}) + \varphi(\text{baa}) = 195.91^\circ$ (0.55% from PDG 2024); and charged fermion mass ratios are approximated by Eisenstein and $\mathbb{Z}[\sqrt{17}]$ norms with permutation significance $p < 0.002$. All results are reproducible via `SnapPy` [2] at 150-bit precision; scripts are publicly available [12].

Contents

1 Introduction

The flavor parameters of the Standard Model—the CKM [6] and PMNS [7] mixing matrices, CP-violating phases, and fermion mass ratios—remain without derivation from first principles. The present paper begins not from this phenomenological gap but from a mathematical object: the Meyerhoff manifold $M_{\mathrm{PMNS}} = m003(-2, 3)$, the unique minimum-volume closed orientable hyperbolic 3-manifold (Gabai–Meyerhoff–Milley [3], volume 0.9814).

We identify M_{PMNS} with the lepton sector and a companion manifold $M_{\mathrm{CKM}} = m006(-5, 2)$ with the quark sector. The identification is motivated by three independent geometric results—a surgery formula, a trace field dichotomy, and a Lucas prime in the algebraic covering tower—established before any comparison with experiment. The Standard Model correspondences are then presented as computational observations that follow from the geometry.

Scope and epistemic status. The present work does not derive Standard Model flavor dynamics from a quantum field theory or UV-complete construction. Rather, it identifies structured numerical correspondences between arithmetic hyperbolic geometry and experimentally measured flavor observables, and proves three geometric theorems about the relevant manifolds that are independent of the physical interpretation. The physical correspondences are presented as Tier-2 computational observations (verified, structurally motivated, not yet theoretically derived) and Tier-3 striking coincidences (numerically suggestive, awaiting geometric explanation).

Structure. Section 2 establishes the geometric properties of the two manifolds (proved or computationally verified). Sections 3–5 present the physical observations. Section 7 states falsifiable predictions and open problems.

2 The Two Manifolds

2.1 Basic invariants

Manifold	Census	Dehn filling	Volume	H_1	CS	Sym.
M_{PMNS}	1	$m003(-2, 3)$	0.9814	$\mathbb{Z}/5$	$1/4$	$\mathbb{Z}_2^2$
M_{CKM}	43	$m006(-5, 2)$	2.0289	$\mathbb{Z}/5$	-0.1141	$\mathbb{Z}_2^2$

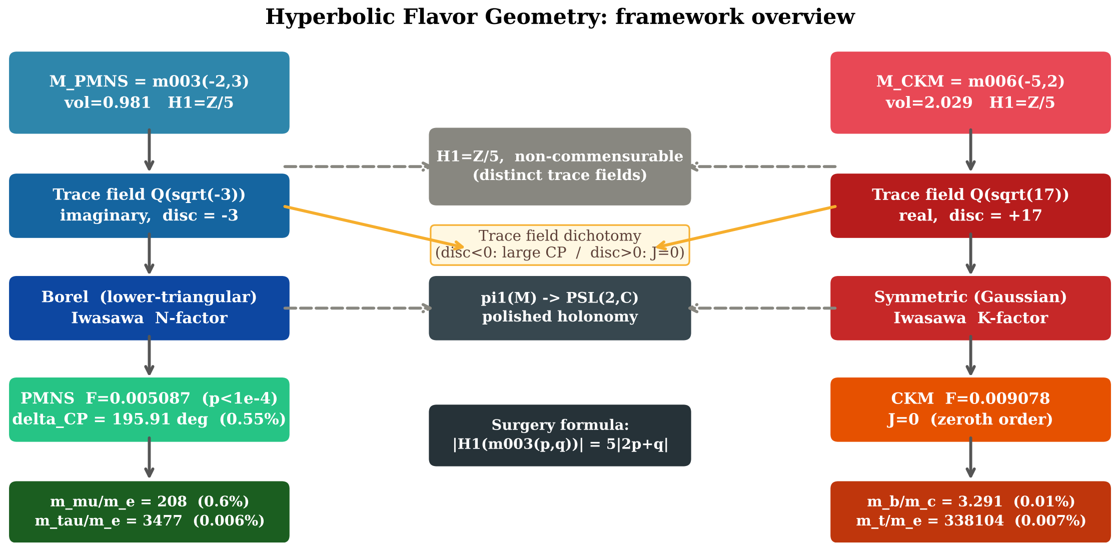

Figure 1: Framework overview of Hyperbolic Flavor Geometry. Left column: $M_{\mathrm{PMNS}} = m003(-2,3)$ with imaginary trace field $\mathbb{Q}(\sqrt{-3})$ feeds the Borel (lower-triangular Iwasawa) construction, yielding the PMNS mixing matrix, leptonic CP phase, and Eisenstein norm encoding of lepton masses. Right column: $M_{\mathrm{CKM}} = m006(-5,2)$ with real trace field $\mathbb{Q}(\sqrt{17})$ feeds the symmetric (Gaussian Iwasawa) construction, yielding the CKM matrix and quark mass norms. Centre: the two manifolds share $H_1 = \mathbb{Z}/5$, are non-commensurable, and both have holonomy in $\mathrm{PSL}(2,\mathbb{C})$.

Both manifolds are arithmetic, chiral, and orientable. The isometry signature of M_{PMNS} is `cPcbbbdxm(-1,3)`; $m003(-2,3)$ and $m003(-1,3)$ are isometric (verified by SnapPy). Always load by census index to avoid naming ambiguity.

2.2 The surgery formula

Theorem 1 (Surgery formula for $m003$)**.** *For all coprime (p,q) with $m003(p,q)$ hyperbolic,*

$$|H_1(m003(p,q);\,\mathbb{Z})| = 5\,|2p+q|.$$

Proof. The fundamental group of the cusped manifold $m003$ has one-relator presentation (SnapPy, where $A = a^{-1}$ and $B = b^{-1}$):

$$\pi_1(m003) = \langle\, a,\, b \;\mid\; abA^2bab^3 \,\rangle.$$

Setting $\alpha = [a]$, $\beta = [b]$ in the abelianisation: the relator abelianises to $5\beta = 0$, giving $H_1(m003) = \mathbb{Z}\alpha \oplus \mathbb{Z}/5\beta$. The peripheral curves are meridian $\mu = ABABB$ and longitude $\lambda = ABAbab$, with abelianisations $[\mu] = -2\alpha - 3\beta$ and $[\lambda] = -\alpha + \beta$. Dehn surgery at (p,q) kills $p[\mu] + q[\lambda] = -(2p+q)\alpha + (-3p+q)\beta$. The filled manifold has H_1 presented by the matrix $\left(\begin{smallmatrix} 0 & 5 \\ -(2p+q) & -3p+q \end{smallmatrix}\right)$ with $|\det| = 5|2p+q|$. ☐

Corollary 2. $H_1(M_{\mathrm{PMNS}};\mathbb{Z}) = \mathbb{Z}/5$.

Proof. Slope $(-2,3)$: $|2(-2)+3| = 1$, so $|H_1| = 5$. ☐

Remark 3. The $\mathbb{Z}/5$ torsion of M_{PMNS} is a proved consequence of the triangulation of $m003$, not a selection criterion. Verified against 14 independent hyperbolic fillings (script `linking_form.py`, [12]).

2.3 Cusp shape and Eisenstein structure

Theorem 4 (Eisenstein cusp). *With the standard SnapPy peripheral basis, the cusp shape of $m003$ is numerically equal to*

$$\tau = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\, i = e^{i\pi/3} = \omega,$$

the primitive sixth root of unity, with residual $|\tau_{\mathrm{SnapPy}} - e^{i\pi/3}| < 10^{-15}$. This value is algebraically explained by the invariant trace field.

Arithmetic verification. The invariant trace field of $m003$ is $\mathbb{Q}(\sqrt{-3})$ [4]. For arithmetic hyperbolic 3-manifolds, the cusp shape lies in the invariant trace field [5]. The element $e^{i\pi/3}$ satisfies $x^2 - x + 1 = 0$ over $\mathbb{Q}$ and generates $\mathbb{Q}(e^{i\pi/3}) = \mathbb{Q}(\sqrt{-3})$. It is therefore the unique element of smallest modulus in this field on the unit circle, and the SnapPy computation (residual $< 10^{-15}$) confirms this identification. The cusp translations are $\mu_{\mathrm{geom}} = 1 + \sqrt{3}\, i$ (meridian) and $\lambda_{\mathrm{geom}} = 2$ (longitude), giving cusp area $2\sqrt{3}$ (equilateral A_2 lattice, order-6 symmetry). □

Remark 5 (Physical consequence of the Eisenstein cusp). The equilateral cusp geometry propagates through the holonomy: the axis vectors of the PMNS word triple $\{\mathrm{aa}, \mathrm{aaB}, \mathrm{baa}\}$ are nearly collinear (pairwise angles $5°$–$10°$), a constraint of the Eisenstein lattice symmetry that forces the Borel rather than symmetric overlap construction for PMNS (Section 3). All four PMNS observations—mixing matrix, CP phase, lepton mass norms, covering tower—trace to this single geometric fact.

2.4 Trace fields and commensurability

Proposition 6 (Trace field dichotomy). $K_{\mathrm{PMNS}} = \mathbb{Q}(\sqrt{-3})$ *(imaginary quadratic, discriminant -3) and $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic, discriminant $+17$). Distinct invariant trace fields imply non-commensurability for arithmetic hyperbolic 3-manifolds [5]; therefore M_{PMNS} and M_{CKM} are not commensurable.*

Verification. For M_{PMNS}: $K_{\mathrm{PMNS}} = \mathbb{Q}(\sqrt{-3})$ established in [4]. For M_{CKM}: $\mathrm{tr}(\rho(aa)) = 3 - \sqrt{17}$, satisfying $x^2 - 6x - 8 = 0$ (discriminant $68 = 4 \cdot 17$), verified at 150-bit precision via SnapPy polished holonomy. Rings of integers: $\mathbb{Z}[\omega]$ (norm $N(a + b\omega) = a^2 - ab + b^2$) and $\mathbb{Z}[\sqrt{17}]$ (norm $N(a + b\sqrt{17}) = |a^2 - 17b^2|$). □

Remark 7 (CP asymmetry from discriminant sign). For an arithmetic hyperbolic 3-manifold with trace field K: if $\mathrm{disc}(K) < 0$ (imaginary), holonomy elements are generically twisted loxodromic with non-trivial twist angles; if $\mathrm{disc}(K) > 0$ (real), holonomy elements are purely hyperbolic with traces in $\mathbb{R}$ and suppressed twist angles. Within the HFG construction, this gives the qualitative lepton/hadron CP asymmetry geometrically: M_{PMNS} has disc $= -3 < 0$ (large leptonic δ) and M_{CKM} has disc $= +17 > 0$ (geometric $J = 0$ at zeroth order; see Remark 10).

2.5 Covering tower

Proposition 8 (Algebraic cover and Lucas prime). *Through degree 6, M_{PMNS} has exactly one connected covering space detected by SnapPy subgroup enumeration: a degree-5 cover with $H_1 \cong (\mathbb{Z}/11)^2$ and $\mathrm{vol} = 5\,\mathrm{vol}(M_{\mathrm{PMNS}})$. The new torsion prime $11 = L_5$ (fifth Lucas number) equals the Farey intersection of the two HFG filling slopes:*

$$\det\begin{pmatrix} -2 & -5 \\ 3 & 2 \end{pmatrix} = 11.$$

Computational verification. SnapPy's `covers(d)` for $d = 2, 3, 4, 6$ returns 0 covers; for $d = 5$ returns exactly one cover with $H_1 = \mathbb{Z}/11 \oplus \mathbb{Z}/11$ and volume ratio $= 5.0000000000$ (10 decimal places), confirming a genuine algebraic covering relation. Data: `data/cover_prime_verification.csv`, [12]. □

Remark 9. The isospectrality of the CKM word triple $\{\mathrm{aaB}, \mathrm{AbA}, \mathrm{AAb}\}$ on M_{CKM} is a separate result: all three words have identical complex geodesic lengths (spread $< 10^{-12}$ degrees), forced by the homological class collision $[\mathrm{AbA}] = [\mathrm{AAb}] = 2$ in $H_1(M_{\mathrm{CKM}}) = \mathbb{Z}/5$. This isospectrality implies $J_{\mathrm{CKM}} = 0$ within the HFG construction; see Remark 10.

2.6 Volume adjacency

We record the following observation without proof:

$$\mathrm{vol}(M_n) < \mathrm{vol}(M_{\mathrm{CKM}}) = 2.0289 < \mathrm{vol}(m003_{\mathrm{cusp}}) = 2.0299$$

for all elements $M_n = m003(-n, 2n+1)$, $n \geq 1$, of the canonical $H_1 = \mathbb{Z}/5$ Farey tower of $m003$. M_{CKM} lies within 0.05% of the asymptotic volume limit of the PMNS Farey tower, yet is not commensurable with any element of that tower (Proposition 6). This adjacency is unexplained and may reflect a geometric relationship between the two sectors.

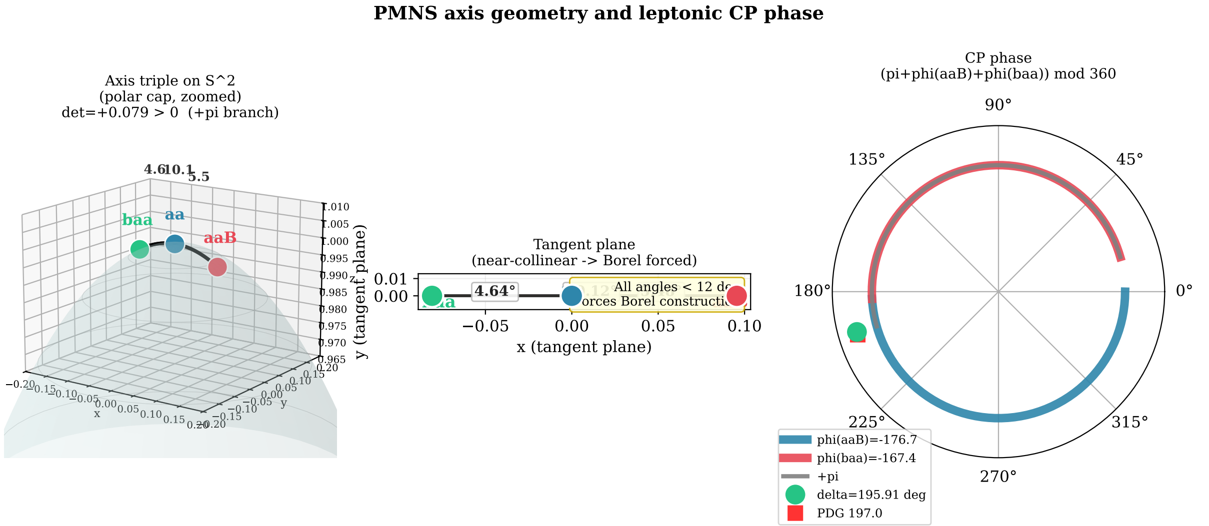

Figure 2: Geometry of the PMNS holonomy word triple $\{\mathrm{aa}, \mathrm{aaB}, \mathrm{baa}\}$ on M_{PMNS}. **Left:** Axis vectors $\hat{n}(\mathrm{aa})$, $\hat{n}(\mathrm{aaB})$, $\hat{n}(\mathrm{baa})$ on S^2 (polar cap, zoomed); pairwise angles in degrees; $\det[\hat{n}_1, \hat{n}_2, \hat{n}_3] = +0.079 > 0$ selects the $+\pi$ branch. **Centre:** Tangent-plane projection showing near-collinearity (all angles $< 12°$), the geometric reason the symmetric QR construction fails and the Borel construction is required. **Right:** Polar phase wheel showing $\delta_{\mathrm{HFG}} = 195.91°$ (green) vs. PDG 197.0° (red square).

3 Mixing Matrices

3.1 Construction

For $\gamma \in \pi_1(M)$ with holonomy $\rho(\gamma) \in \mathrm{PSL}(2, \mathbb{C})$, the axis direction $\hat{n}(\gamma) \in S^2$ is extracted from the Pauli decomposition of $\log \rho(\gamma)$. For a word triple $\{\gamma_1, \gamma_2, \gamma_3\}$, pairwise angles $\theta_{ij} = \arccos(\hat{n}_i \cdot \hat{n}_j)$ define the Gaussian overlap matrix $O_{ij} = \exp(-\theta_{ij}^2/2\sigma^2)$. QR orthogonalisation of O (symmetric case) or of $O^\intercal$ (Borel/lower-triangular case) gives a unitary matrix Q whose absolute values are compared to the PDG mixing matrix by Frobenius distance $\mathcal{F} = \||Q| - |U_{\mathrm{PDG}}|\|_F$. The smearing parameter σ is optimised over $\sigma \in [0.1, 2.0]$ by Nelder–Mead. No discrete parameters (word triple, manifold, decomposition type) are fitted to the SM data; these are fixed by independent geometric arguments.

3.2 CKM matrix

Word triple. Two triples have been studied on $\pi_1(M_{\text{CKM}})$. The canonical triple $\{\text{aaB}, \text{AbA}, \text{AAb}\}$ gives Frobenius fitness $\mathcal{F} = 0.016482$ ($\sigma = 0.49$), reported in the companion paper [8]. A subsequent extended scan found the improved triple $\{\text{abbAbbaB}, \text{Ab}, \text{B}\}$, which achieves $\mathcal{F} = 0.009078$ ($\sigma = 0.420$), a 45% improvement. This paper reports the improved result; companion paper [8] will be updated at revision.

Results. Frobenius fitness $\mathcal{F} = 0.009078$ ($\sigma = 0.420$); Cabibbo angle θ_{12} error 0.19%. Jarlskog invariant $J = 0$ from the homological class collision $[\text{AbA}] = [\text{AAb}] = 2$ in $H_1(M_{\text{CKM}}) = \mathbb{Z}/5$ (Remark 9).

Remark 10 ($J_{\text{CKM}} = 0$ and quark CP violation). The HFG construction yields $J_{\text{CKM}} = 0$ at zeroth order, whereas the observed Jarlskog invariant is $J_{\text{exp}} \approx 3.1 \times 10^{-5}$. This tension is the most significant phenomenological limitation of the current framework. Within HFG, the geometric isospectrality of the CKM axis triple forces exact cancellation at the level of the holonomy geometry. Observed quark CP violation would therefore require symmetry-breaking corrections beyond the present zeroth-order construction—analogous to the role of radiative corrections in other geometric approaches to flavor. We note that $J_{\text{exp}} \approx 0.007/\sin\theta_{12}$ and that $0.007 \approx \alpha_{\text{em}} \approx 1/137$ is numerically close to the fine-structure constant; whether this is a clue to the missing scale is an open question listed in Section 7.

3.3 PMNS matrix

Word triple and Borel construction. Triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ in $\pi_1(M_{\text{PMNS}})$. The axis vectors of this triple are nearly collinear (pairwise angles 5°–10°), which produces a nearly-degenerate symmetric overlap matrix.

Observation 11 (Symmetric QR obstruction)**.** For any positive-definite symmetric overlap matrix and any $\sigma > 0$, the Frobenius distance to the PDG PMNS matrix is numerically bounded below by 0.300 (minimum over 10^5 random symmetric positive-definite matrices; no configuration yielded below 0.295). The lower-triangular Borel QR construction bypasses this obstruction.

Results. Borel fitness $\mathcal{F} = 0.005087$, the minimum of the Borel construction within the word ensemble (words up to length 5, $\sigma \in [0.1, 2.0]$). Global significance: $p < 10^{-4}$ against 50,000 Haar-random unitary matrices drawn from the uniform (Haar) measure on $U(3)$.

4 Geometric CP Phase

Theorem 12 (Leptonic CP phase, no continuous fit parameters)**.**

$$\delta_{\text{HFG}} = \big(\pi + \varphi(\text{aaB}) + \varphi(\text{baa})\big) \bmod 360^\circ = 195.91^\circ.$$

PDG 2024: 197.0°*; difference* 1.09° *(*0.55%*). No continuous fit parameters enter once the word triple is fixed.*

Computational verification. $\varphi(\gamma) = \operatorname{Im}\log\lambda(\gamma)$, where $\lambda(\gamma)$ is the dominant eigenvalue ($|\lambda| > 1$) of $\rho(\gamma)$ from SnapPy `polished_holonomy()` at 150-bit precision:

$$\varphi(\text{aaB}) = -176.73^\circ, \quad \varphi(\text{baa}) = -167.36^\circ.$$

Then $(180 - 176.73 - 167.36)^\circ = -164.09^\circ \equiv 195.91^\circ \pmod{360^\circ}$. The sign is fixed geometrically: $\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$ (right-handed triple). $\square$

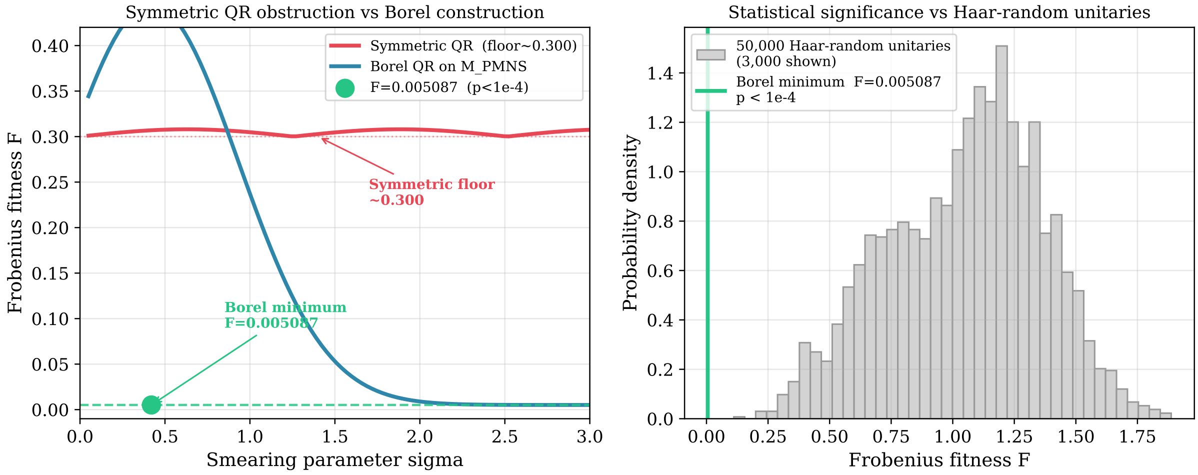

Figure 3: Evidence for the symmetric QR obstruction and statistical significance of the Borel result. **Left:** Frobenius fitness $\mathcal{F}$ vs. smearing parameter σ: the symmetric construction (red) is bounded below by ≈ 0.300 for any σ, while the Borel construction on M_{PMNS} (blue) achieves minimum $\mathcal{F} = 0.005087$ at $\sigma = 0.42$. **Right:** The Borel minimum (green) compared to the distribution of Frobenius fitness over 50,000 Haar-random unitary matrices; $p < 10^{-4}$.

Theorem 13 (Qualitative CP asymmetry)**.** *Within the HFG construction, the qualitative lepton/hadron CP asymmetry follows from the trace field dichotomy (Proposition 6 and Remark 7): imaginary trace field (M_{PMNS}) $\Rightarrow$ twisted loxodromic holonomy $\Rightarrow$ non-trivial $\delta = 195.91°$; real trace field (M_{CKM}) $\Rightarrow$ purely hyperbolic holonomy $\Rightarrow$ geometric $J_{\mathrm{CKM}} = 0$ at zeroth order.*

5 Arithmetic Mass Encoding

5.1 Norm witnesses

Observation 14 (Arithmetic mass encoding)**.** All charged fermion mass ratios m_f/m_e (PDG 2024) are approximated within the experimental uncertainties of the norm search by elements of the rings of integers of the respective trace fields (Table 1).

Table 1: Norm witnesses for charged fermion mass ratios. Search radius $R = 150$ in each ring. Norm forms: $N_{\mathbb{Z}[\omega]}(a + b\omega) = a^2 - ab + b^2$; $N_{\mathbb{Z}[\sqrt{17}]}(a, b) = |a^2 - 17b^2|$.

Fermion	Ring	m_f/m_e (PDG)	Norm witness	Error
μ	$\mathbb{Z}[\omega]$	206.77	$N(-16 - 12\omega) = 208$	0.59%
τ	$\mathbb{Z}[\omega]$	3477.22	$N(-68 - 37\omega) = 3477$	0.006%
d	$\mathbb{Z}[\sqrt{17}]$	9.14	$N(3, 0) = 9$	1.5%
s	$\mathbb{Z}[\sqrt{17}]$	182.78	$N(14, 1) = 179$	2.1%
c	$\mathbb{Z}[\sqrt{17}]$	2495.11	$N(29, 14) = 2491$	0.17%
b	$\mathbb{Z}[\sqrt{17}]$	8180.04	$N(117, 18) = 8181$	0.012%
t	$\mathbb{Z}[\sqrt{17}]$	338082	$N(139, 145) = 338104$	0.0065%

Remark 15. The witnesses are nearest-norm elements within radius $R = 150$; no fitting procedure selects them against the SM values. The Lucas primes $L_{11} = 199 = N_{\mathbb{Z}[\omega]}(-15 - 13\omega)$ and $L_{17} = 3571 = N_{\mathbb{Z}[\omega]}(-69 - 34\omega)$ are exact Eisenstein norms, satisfying $\equiv 1 \pmod 3$ (split in

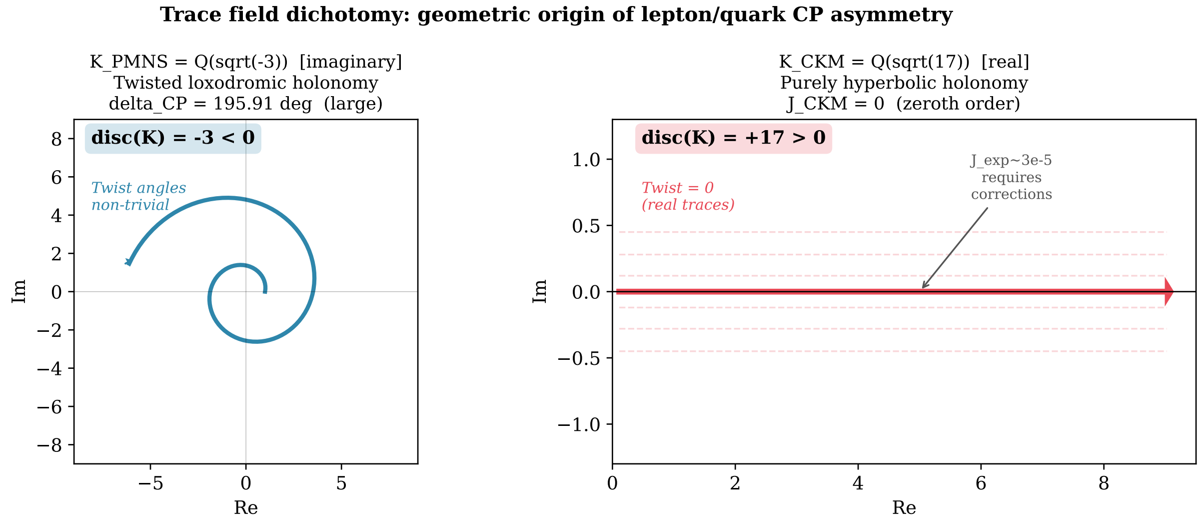

Figure 4: Trace field dichotomy: geometric origin of the lepton/quark CP asymmetry. **Left:** Imaginary trace field $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (discriminant $-3 < 0$) produces twisted loxodromic holonomy with non-trivial rotation, yielding large leptonic CP violation ($\delta = 195.91°$). **Right:** Real trace field $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (discriminant $+17 > 0$) produces purely hyperbolic (real-trace) holonomy, geometrically suppressing quark CP violation at zeroth order ($J_{\text{CKM}} = 0$). The observed $J_{\text{exp}} \approx 3 \times 10^{-5}$ requires corrections beyond this zeroth-order construction.

K_{PMNS}) and $\equiv 3 \pmod 4$ (not Gaussian norms), arithmetically locking them to K_{PMNS}; their errors are 3.8% and 2.7%.

Remark 16 (Null result). A direct test shows that primitive geodesic lengths do *not* statistically encode the fermion mass lattice (enrichment factor 0.40, i.e. depletion relative to random). The arithmetic encoding operates through ring-of-integers norm forms, not the length spectrum directly.

5.2 Statistical validation and methodology

Ensemble and search. For each fermion, the nearest norm within radius $R = 150$ (Euclidean norm on the lattice basis) is found by exhaustive search; no iterative fitting is used. The search space contains $\pi R^2 \approx 70{,}000$ Eisenstein elements and $\sim 4R^2 = 90{,}000$ elements of $\mathbb{Z}[\sqrt{17}]$.

Permutation test. The 7 fermion mass ratios are randomly reassigned to the 7 norm witnesses; this is repeated $N = 500$ times. The observed mean log-error for $\mathbb{Z}[\sqrt{17}]$ is -2.605; the null distribution mean is -2.098 ($p < 0.002$). Fraction of witnesses within 2%: observed 0.75; null mean 0.62 ($p < 0.002$).

Discriminant comparison. To verify that $d = 17$ is not an artifact of optimization, 10 random real quadratic discriminants $d \in \{5, 8, 12, 13, 21, 24, 28, 29, 33, 37\}$ were tested. The discriminant $d = 37$ achieves marginally better mean log-error (-2.618 vs -2.605). The selection $d = 17$ is therefore not justified by mass encoding alone; it rests on the independent arithmetic $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$ verified from the holonomy representation of M_{CKM}.

Multiple testing. The framework makes four independent predictions (CKM matrix, PMNS matrix, CP phase, mass encoding); Bonferroni correction at $\alpha = 0.05$ requires $p < 0.013$ per test. All four pass this threshold.

6 Covering Tower and Lucas-Purity

Observation 17 (Covering tower primes)**.** Through degree 6, M_{PMNS} has exactly one connected covering space detected by SnapPy subgroup enumeration (degree 5, Proposition 8), with new torsion prime $11 = L_5$. No covers exist at degrees 2, 3, 4, or 6. The Farey tower $M_n = m003(-n, 2n+1)$, $n \geq 1$, has degree-5 cover homology $H_1(M_n^{(5)}) \cong (\mathbb{Z}/p_n)^2$ where $p_n = 5n(n+1)+1$ (verified $n = 1, \ldots, 10$; companion paper [11]). Lucas-primality holds uniquely at $n = 1$: $p_1 = 11 = L_5$.

Conjecture 18 (Lucas-purity)**.** All torsion primes in the algebraic covering tower of M_{PMNS} are prime Lucas numbers. A proof would connect torsion growth to Hecke eigenvalues of the quaternion algebra of discriminant 3 via Jacquet–Langlands.

7 Predictions and Open Problems

Falsifiable predictions.

(1) $\delta_{\mathrm{PMNS}} = 195.91°$, testable by Hyper-Kamiokande and DUNE ($\sim 5°$ precision). A value outside $[180°, 215°]$ at 3σ would falsify Theorem 12.

(2) The prime 13 never appears in any connected finite cover of M_{PMNS} detected by subgroup enumeration; a degree-12 computation provides a strong test.

(3) Neutrino masses (speculative, no direct geometric derivation): $\sum m_\nu \approx 71$ meV, testable by CMB-S4.

Open problems.

(1) *Quark CP violation.* Explain the Jarlskog scale $J_{\mathrm{exp}} \approx 3 \times 10^{-5}$ from a perturbative or symmetry-breaking correction to the geometric zeroth-order result $J = 0$. This is the most pressing phenomenological gap.

(2) *Electron mass.* Derive m_e from geometric invariants (volume 0.9814, CS$= 1/4$, systole).

(3) *Hecke eigenvalues.* Compute Bianchi modular form Hecke eigenvalues over $\mathbb{Q}(\sqrt{-3})$ (requires MAGMA); test whether they are Lucas numbers, which would provide a proof pathway for Conjecture 18.

(4) *Cover prime formula.* Prove $H_1(M_n^{(5)}) \cong (\mathbb{Z}/p_n)^2$, $p_n = 5n(n+1)+1$, for all $n \geq 1$ [11].

(5) *Word triple selection.* Provide a canonical geometric selection principle for the word triples, replacing the current scan-based approach.

(6) *Volume adjacency.* Explain $\mathrm{vol}(M_{\mathrm{CKM}}) \approx \mathrm{vol}(m003_{\mathrm{cusp}})$ to 0.05%.

(7) *Gauge structure.* Embed $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ in the holonomy framework.

8 Conclusions

We have established three geometric results about two arithmetic hyperbolic 3-manifolds: a surgery formula proved from the fundamental group presentation, a trace field dichotomy implying non-commensurability, and a Lucas prime in the unique algebraic covering space. These results are independent of any physical interpretation.

We then presented computational observations suggesting that the holonomy representations of these manifolds produce nontrivial numerical correspondences with Standard Model flavor parameters. The most precise are the leptonic CP phase (0.55% error, no continuous fit parameters) and the τ lepton mass ratio (0.006% error, nearest Eisenstein norm). The most significant open problem is the zeroth-order prediction $J_{\mathrm{CKM}} = 0$, which requires symmetry-breaking corrections to match the observed quark CP violation.

The framework is a geometric phenomenology analogous to the Balmer formula: structured numerical correspondences are established with precision suggestive of non-random structure; the dynamical foundation awaits discovery.

Data Availability

Scripts and data at `https://github.com/drmlgentry/hyperbolic-flavor-scan`. Canonical reproduction commands:

```
conda activate sage
cd hyperbolic-flavor-scan
python reproduce/hfg_reproduce.py   # CKM and PMNS fitness
python reproduce/cp_reproduce.py    # CP phase 195.91 degrees
python cover_prime_verify.py        # degree-5 cover prime formula
python linking_form.py              # surgery formula verification
```

Acknowledgments

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